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Design and Implementation of a continuous A/B testing framework using sequential testing methods

Study Aim

- Investigate how peeking and early stopping affect the False Positive Rate (FPR) and statistical power in A/B testing.
- Evaluate sequential testing methods that support continuous monitoring while controlling FPR and maintaining statistical power.
- Identify methods to reduce sample size and accelerate experimental decisions without compromising statistical reliability.

CTR in A/B testing

CTR = No. of Clicks/ No. of Impression

According to CLT (Central Limit theorem) if we take repeat samples from a CTR distribution (Bernoulli Trials) it will tend to average around true CTR rate with a Normal Distribution

For testing two advertising campaigns based on Click-Through Rate (CTR), use the following standard two-sided hypotheses:

Effect Of Sample Size

$$N = \frac{2\sigma^2\left(z_{1-\frac{\alpha}{2}} + z_{1-\beta}\right)^2}{d_{min}^2}$$

A/B Testing

A/B testing is an experimental method used to compare two versions (A and B) of a product, webpage, or advertising campaign to determine which performs better based on a specific metric, such as CTR or conversion rate.

Peeking in A/B Testing

Peeking in A/B testing is the practice of repeatedly checking experimental results before the planned sample size is reached.

Early stopping in A/B testing is the practice of ending an experiment before reaching its originally planned sample size or duration because a predefined stopping condition has been met.

FPR in A/B Testing

False Positive Rate (FPR) in A/B testing is the probability of declaring a statistically significant difference between versions A and B when, in reality, there is no true difference.

In an A/A test, both groups receive the same treatment. Therefore, there should be no real difference between their performance, and any statistically significant difference is a false positive, assuming the test's null hypothesis is true.

Power Analysis in A/B Testing

Power analysis in A/B testing determines the sample size needed to detect a real difference between versions A and B with a specified probability.

- Statistical power: Probability of detecting a real effect (commonly 80%).
- Significance level : Acceptable Type I error rate (commonly 5%).
- Minimum Detectable Effect (MDE): Smallest effect you want to detect.
- Sample size: Number of users or impressions required per group

SPRT

SPRT (Sequential Probability Ratio Test) is a statistical method used in A/B testing that continuously evaluates incoming data and allows an experiment to stop early when sufficient evidence supports one hypothesis over another.

- H_0: No meaningful difference between versions A and B.
- H_1: A meaningful difference exists.
- Decision: Continue collecting data, accept evidence for H_0, or accept evidence for H_1 based on predefined decision boundaries.

Theory [edit]

As in classical hypothesis testing, SPRT starts with a pair of hypotheses, say H_0 and H_1 for the null hypothesis and alternative hypothesis respectively. They must be specified as follows:

$$H_0 : p = p_0$$
$$H_1 : p = p_1$$

The next step is to calculate the cumulative sum of the log-likelihood ratio, $\log \Lambda_i$, as new data arrive: with $S_0 = 0$, then, for i=1,2,...,

$$S_i = S_{i-1} + \log \Lambda_i$$

The stopping rule is a simple thresholding scheme:

- $a < S_i < b$: continue monitoring (*critical inequality*)
- $S_i \geq b$: Accept H_1
- $S_i \leq a$: Accept H_0

where a and b ($a < 0 < b < \infty$) depend on the desired type I and type II errors, α and β. They may be chosen as follows:

$$a \approx \log \frac{\beta}{1-\alpha} \text{ and } b \approx \log \frac{1-\beta}{\alpha}$$